

An action research to develop critical thinking skills in the context of citizenship education in higher education

Arzu Saldıray^{a,*}, Ahmet Doğanay^b

^a Nevşehir Hacı Bektaş Veli University, Faculty of Education, Nevşehir, Türkiye

^b Çukurova University, Faculty of Education, Adana, Türkiye

ARTICLE INFO

Keywords:

Critical thinking skills
Critical citizenship approach
Distinguishing the argument
Writing pedagogy
Online group discussions

ABSTRACT

This article aims to demonstrate how critical thinking skills can be developed as a citizenship competence in higher education. Critical thinking skills are necessary for citizens to participate in democratic processes effectively and consciously. The participants of this “action research” based study are the researcher who conducts the undergraduate course Contemporary World Issues and 30 preservice social studies teachers taking this course. Action research was conducted through distance education at a university during the COVID-19 pandemic. Data were collected through Zoom lessons, student products, semi-structured interviews, semi-structured focus group interviews, and WhatsApp text conversations. Descriptive and content analyses were conducted using MAXQDA 2020 software. According to the research findings, a curriculum based on distinguishing arguments, writing pedagogy, and online group discussions improved students’ critical thinking skills. Students used distinguishing arguments as a cognitive framework for solving social and political problems and thus developed citizenship competence. Throughout the process, feedback was found to be quite functional. The curriculum used in the action research process can be recommended for use in the development of critical thinking skills as a citizenship competency in higher education.

1. Introduction and context

With an increase in international migration, population movement, and globalization, heterogeneous societies have developed multi-ethnic and multicultural characteristics, requiring the consideration of citizenship education from new perspectives (Darmanin & European Students’ Union, 2020). While modern society is progressing toward individualization, life lived within the framework of rules is moving away from being linear, and life itself has come to involve questioning what the rules are (Veugelers, 2007). Individualization, in which individuals are expected to make increasingly many choices in a globalized world, continues to clash with globalization. Current pedagogical practices associated with citizenship in educational institutions must be addressed and implemented to respond to contemporary concerns. The contribution of civic education and critical thinking skills is more crucial than ever these days when complexity and continuous uncertainty are experienced more intensely in everyday life as a result of the COVID-19 pandemic (International Civic and Citizenship Education Study, (2022).

There is widespread acceptance of the necessity and importance of critical thinking regarding citizens’ participation in democracy

This study is based on the findings of the first author’s doctoral dissertation prepared under the supervision of the second author.

* Corresponding author at: Department of Educational Sciences, Nevşehir Hacı Bektaş Veli University, 50300, Nevşehir, Turkey.

E-mail address: arzusaldiray@nevsehir.edu.tr (A. Saldıray).

<https://doi.org/10.1016/j.tsc.2024.101584>

Received 13 April 2023; Received in revised form 26 June 2024; Accepted 27 June 2024

Available online 30 June 2024

1871-1871/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Brookfield, (1987); Johnson and Morris, (2010); Lipman, (2003); Ten Dam and Volman, (2004); Volman & Ten Dam, (2015). Through critical thinking, citizens can make critical and conscious contributions to their society. Critical thinking requires reliable investigative procedures rather than careless or distorted thinking. Individuals who act without questioning the reasoning behind their actions, who do not question their beliefs and who, therefore, do not have a say in their lives, cannot be free. (Siegel 1988) states that critical thinking is a tool used by individuals to achieve self-efficacy. People who exhibit self-efficacy are free of irrelevant, unintended, or unjustifiable beliefs.

The development of critical thinking skills is essential to economic productivity Alvarez-Huerta et al., (2022); Arum and Roksa, (2011). According to a report prepared by the (World Economic Forum 2020) on the professions and skills of the future in light of the global economic situation and technological development, self-management skills, including critical thinking and analysis, problem solving and active learning, resilience, stress tolerance, and flexibility, are among the skills employers anticipate to emerge in leadership by 2025. According to (Johnson and Morris 2010), a more innovative, independent, creative, and reflective workforce is the key to increasing a country's competitiveness in the global economy. To address this issue, many countries have incorporated critical citizenship forms into their curricula.

The development of critical thinking skills is a major objective of higher education Arum and Roksa, (2011); Lau, (2015). Education received at the university level plays a particularly important role, as it marks the final step in the transition to professional life and is also a critical period for socialization and individualization.

In the context of studies that emphasize civic competence as critical thinking skills, previous studies (Gürkaynak, 2016; Sönmez, 2016) of the development of critical thinking skills in students did not address citizenship education, and citizenship education studies (Altuntaş, 2016; Kansu, 2015) did not address thinking skills. As well as studies focusing on critical thinking skills within the context of civic competencies (Acun et al., 2010), correlational studies have been conducted in the screening model to assess critical thinking orientations and democratic values Sivacı and Altaş, (2023). Research on critical thinking skills in citizenship education (Çayır, 2014; Gürkaynak et al., 2005) has mostly been published in books. When international studies on the subject are examined, it is seen that teachers have different conceptions of citizenship, such as critical multicultural citizenship (Castro, 2009; Girón, 2012), democratic citizenship (Ververi, 2012), active citizenship (Gonsalvez, 2013), and global citizenship Pugliese, (2015). Moreover, empirical studies have been conducted on the effectiveness of methods such as dialogue (Robertson and Rane-Szostak, 1996), group problem solving (Vijayarajnam, 2012), and the constructivist teaching model (Stanley, 2017) on the basis of developing critical thinking. Critical thinking skills have been identified as civic competencies in higher education only a few times (Awan et al., 2018; Piedade et al., 2020; Ten Dam and Volman, 2004) and these studies have had a limited scope. There is a lack of practice-based research on the integration of civic education and thinking skills. The present study aimed to contribute to the limited theoretical literature on citizenship competence from a critical thinking perspective. Additionally, the study was intended to examine the action research process based on distance education in detail and explain the difficulties encountered and their solutions based on empirical findings. This study was also intended to create an application-based curriculum for higher education programs.

This action research aimed to develop higher education students' critical thinking skills as a citizenship competence and analyzed (i) how a change/transformation was achieved in students' critical thinking skills, and (ii) the problems encountered during the implementation process.

Table 1
Theoretical framework of critical citizenship education.

	Politics/Ideology	Social/Collective	Self/Subjectivity	Praxis/Action
Knowledge	To understand and know the emerging relations of history, societies, and systems in terms of oppression, injustice, and power structures	Understanding the relationship between culture, power, and transformation without being swayed by ideas and writings that support the dominant course	Self-perception: Understanding of context, culture, and one's own position	Knowledge of behavior that affects society and injustice, of power, and of being collective for a systematic change
Skill	Developing a policy using concepts of critical and structural social analysis, culture, knowledge, and power, as well as analyzing causality in depth	Thinking holistically, critically interpreting others' points of view, interaction, collaboration, and dialogue skills	Independent critical thinking by speaking with one's own voice, the capacity to think critically about the individual's position in society	Active participation and critical thinking skills, collective action skills that challenge the status quo with the dream of a better world
Value	Commitment to values against inequality and tyranny	Inclusive relationship based on dialogue with the identities and values of others	Being concerned with social justice and paying regard to self-worth	Thinking deeply, having awareness of moral action, and responsibility
Disposition	Actively curious, critically interested in society and public issues, taking action against tyranny and injustice	Social awareness, cooperation, responsible to self and others, willingness to learn with others	Critical perspective; autonomy; responsibility in thought, emotion and action; forward- thinking; in touch with reality	Willingness and dedication to change society, civic courage, the feeling of responsibility for decisions and actions

(Johnson and Morris, 2010, p. 91).

2. Theoretical framework

2.1. Understanding of critical citizenship

When reviewing the literature on citizenship and civic education, one may note that some pedagogical approaches are based on personal freedom, whereas others emphasize social harmony or equality, and each highlights different qualities of a good citizen on the basis of the respective approach. The understanding of “democratic citizenship,” for instance, emphasizes the ability to think critically in order to make conscious and democratic decisions (Darmanin and European Students’ Union, 2020), whereas citizens are expected to have the knowledge, skills, and values determined by the understanding of “active citizenship” (National Council for the Social Studies, 2001), which requires the active participation of an individual possessing the knowledge, skills, and values that make up active citizenship (The Council of the European Union, 2019). Accordingly, political participation is fundamental to active citizenship. On the other hand, the “critical citizenship” approach emphasizes the importance of active participation and having a critical perspective in modern democratic societies Burbules & Berk, (1999); Piedade et al., (2020); Volman and Ten Dam, (2015).

Critical citizenship, which is grounded in critical pedagogy, emphasizes the importance of equipping individuals with an independent and critical perspective. According to this approach, to develop an understanding of citizenship that thinks and acts, education and society must be analyzed in terms of the relationships between ideological, political, and economic fields, as well as the role of schools in all of these Apple, (2006); Giroux, (2014). Through an analysis and comparison of theories and approaches in critical pedagogy and critical thinking, (Johnson and Morris 2010) developed a framework for developing the form of critical citizenship. This theoretical framework is a useful and flexible model for critical citizenship, and is conducive to the development of thinking skills.

Table 1

2.2. Development of critical thinking skills

Developing critical thinking skills does not depend on any inherently progressive process, but rather on a process that humans can develop in many different ways Kuhn, (1999); Perkins, (1987). Research has shown that teachers primarily use three distinct methodologies when developing and evaluating critical thinking skills: (1) oral and written reflection, (2) argumentation, and (3) studies of reading, analyzing, and synthesizing sources Bezanilla et al., (2019).

2.2.1. Argumentation

According to (Andrews 2015), critical thinking is closely related to argumentation, and both critical thinking and argumentation are pedagogical acquisitions for higher education. This is mainly because of the possibility of concretizing argumentation. Different argumentation models can be adapted for writing, speaking, and composition.

In (Cottrell’s 2005) method for identifying arguments among strategies for developing critical thinking skills, arguments and other elements occur in an integrated setting, which can be written or spoken. First, (Cottrell 2005) stated that it is necessary to identify the arguments and supporting evidence within written or spoken materials. Furthermore, it is necessary to determine whether the material includes different perspectives. In the second stage, the other elements intertwined with the argument must be distinguished. These items include the introduction, background information, description, explanation inference, and summary.

2.2.2. Writing

As (Giroux 1978) points out, poor writing reflects poor thinking, and what teachers usually view simply as an “error” in writing is, in fact, a reflection of an error in thinking itself. Writing, therefore, should be considered a dialectical process rather than an instrumental skill (Sheils, 1975, as cited by Giroux, (1978). In other words, writing is an epistemology, a way of learning.

According to (Barnet and Bedau 2011), writing is actually rewriting; that is, revising, and revising is a revision, a second look. (Tsui’s 2002) study of effective pedagogies that promote critical thinking noted that writing and rewriting activities after feedback encourage students to think more deeply about their written products, which is associated with the development of critical thinking skills. (Manalo and Sheppard 2016) examined critical evaluation, another aspect of critical thinking, in university students’ written reports. According to the findings of this study, training in appropriate language use improves students’ critical evaluation skills in written work. In other words, students who received training in academic discourse effectively reflected critical evaluation competencies in their reports.

2.2.3. Classroom discussion

Working in groups is one of the most important learning paths to high-level mental development. As (Kuhn 1999) pointed out, thinking is a process of reason-based argumentation with a social dimension. This thinking process is intertwined with individual beliefs.

In the process of constructing meaning, it is necessary for individuals to communicate with others and thereby take responsibility for sharing and learning Manalo, (2020). Therefore, engaging in dialogue is necessary not only for effective communication but also for an active contribution to the construction of knowledge.

According to (Phillipson and Wegerif 2020), engaging in dialogue is in a sense the act of “thinking together.” Engaging in dialogue involves interacting with different perspectives, trying to understand them, responding to them, being sensitive to differences, and creating new perspectives in the face of these differences. Engaging in dialogue thus involves seeking a common understanding and meaning and entering into a relationship based on mutual respect. Such a collective action approach develops not only group thinking,

but also individual thinking [Phillipson and Wegerif, \(2020\)](#).

[\(Brookfield 2012\)](#) pointed out that critical thinking is most effectively experienced among peers, which is a crucial component of the social learning process, and argumentation is perhaps the most social of all teaching methods. [\(Brookfield 2012\)](#), who proposed a critical discussion method for developing critical thinking, noted that critical discussions include describing assumptions, evaluating evidence, determining inference steps, and building multidimensional perspectives.

3. Method

3.1. Study design

According to [\(Patton 2014, p. 188\)](#), conscious citizens are essential to a healthy and strong democracy, and they need to learn thinking processes to become conscious citizens. Action research has the advantage of being a practical form of inquiry characterized by “a spiral of planning, acting, observing and reflecting” [\(Kemmis, 1993, p. 178\)](#) and by transforming educational practice into participatory democratic educational research, which is in complete parallel with this study’s objectives. A qualitative research design, action research, was used in this study to understand the process based on distance education practices developed in accordance with action research on critical thinking skills, as well as to improve practices and interventions through critical reflection.

Action research can take many forms based on different approaches. In this study, the “liberating/developing/critical” approach, which is one of the types of action research suggested by [Berg \(2001, p. 187\)](#), was used.

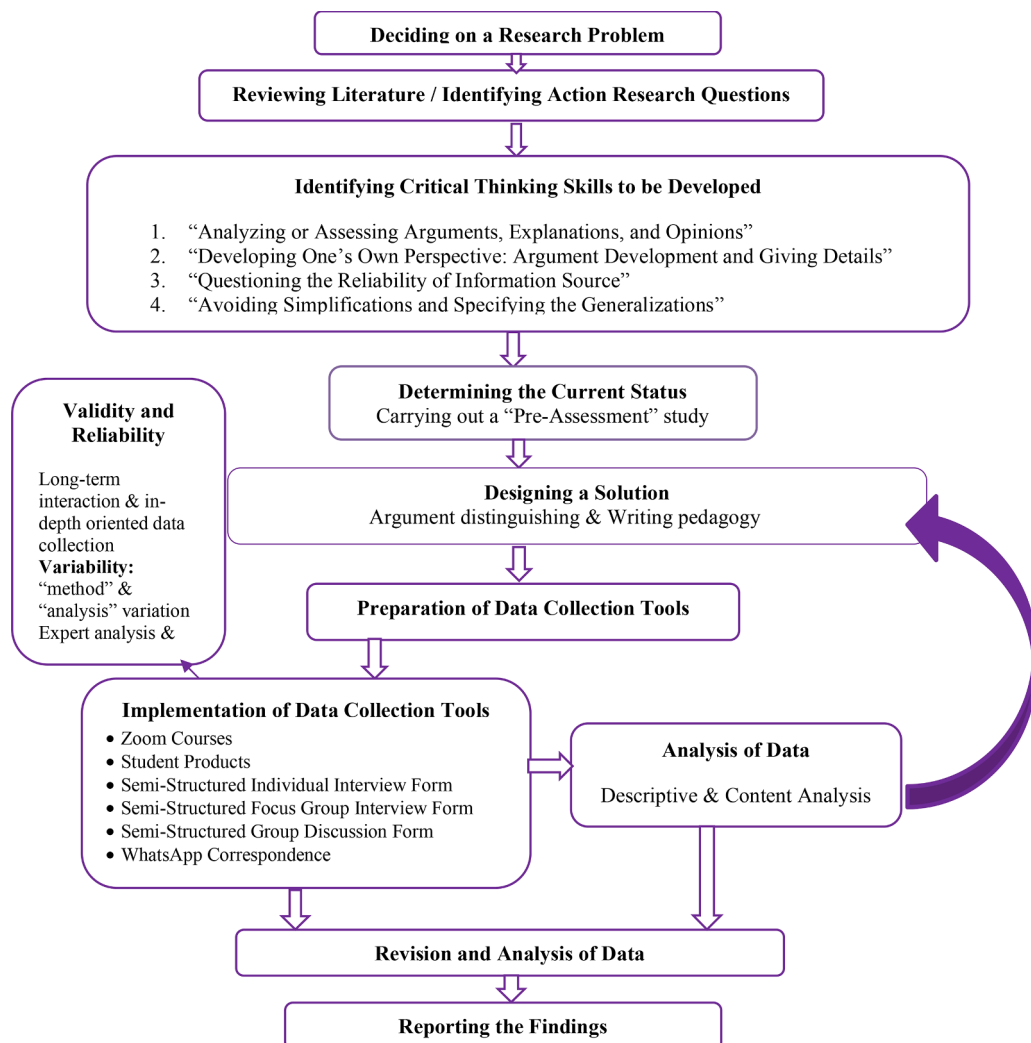


Fig. 1. Study design and the process followed.

3.1.1. Process of action research

The design of the study and the process followed were presented in Fig. 1.

For the purpose of determining the critical thinking skills that were to be developed, a course called Contemporary World Problems

Table 2
Teaching schedule for critical thinking skills

Week	Subject	Critical thinking skills	Activities/ assessment and evaluation
Preparation October 6–20, 2020	A synopsis of world problems	Importance of critical thinking	§ Performing a pre-assessment study
1st Week October 27, 2020	Epidemics and infectious diseases	§ Identifying the argument	1st Activity § Worksheet: “finding the argument” : Determining whether short paragraphs prepared within the scope of the topic contain an argument
2nd Week November 3, 2020	Human environment and its problems	§ Analyzing or assessing arguments, explanations, and opinions § Developing one's own perspective: Argument development and giving details	2nd Activity § Worksheet: “the form of identifying the argument from the other material” Identifying the argument and analyzing the other material in the essay “ <u>an extremely easy prophecy</u> ” § “listening and writing form” : this form should be filled out while listening to the lecturer.
3rd Week November 10, 2020	Population problems	§ Analyzing or assessing arguments, explanations, and opinions § Developing One's Own Perspective: Ability to reflect on the problem by considering the subject from different perspectives	3rd & 4th Activities § Worksheet: “the form of identifying the argument from the other material” Identifying the argument and analyzing the other material in the tedx talk on “ <u>the case for optimism</u> ” by epidemiologist larry brilliant, who played a major role in the eradication of smallpox.
4th Week November 17, 2020	Problem of immigration	§ Developing One's Own Perspective: Ability to reflect on the problem by considering the subject from different perspectives	§ “Listening and writing form” : before completing the form, the student should listen to the tedx talk.
5th Week November 29, 2020	Mid-term exam	General assessment	5th Activity “ The form of identifying the argument from the other material ” § Identifying the argument and analyzing the other material in the essay titled “ <u>the ghost of malthus.</u> ” § Writing : Writing an argument-based essay that reveals one's point of view in response to an open-ended question about current world issues.
6th Week December 2, 2020	Problem of immigration	“ Questioning the reliability of the information source ”	6th Activity § Worksheet: “questioning the reliability of the information source” Turkey's integration policy toward refugees is discussed in two newspaper reports from different perspectives: “ <u>Syrians in Turkey: Is integration possible?</u> ” / “ <u>Integration is OK, now is the time to look forward to the future.</u> ” Students are asked to analyze both articles according to the assessment criteria in the worksheet. Next, they are instructed to compare the two news stories and show the similarities and differences, and then they are instructed to conclude by writing which news story they find persuasive with evidence.
7th Week December 8, 2020	Problem of refugees	“ Avoiding simplifications and specifying the generalizations ”	7th Activity § Worksheet: “the form of identifying the argument from the other material” § Identifying the argument and analyzing the other material in the essay “In the Supermarket.” Finding expressions of generalization and stereotyping in the essay titled “In the supermarket.” § Writing : The students are asked to write an essay that evaluates the positive and negative results of the recent refugee movement in Turkey.
8th Week December 15, 2020	Refugee problem	All	8th Activity 1st Group discussion : a group discussion (zoom meeting) is held with the participation of 9 students who have chosen the topic of “refugee problem.”
9th Week December 26, 2020	Women's problems	All	9th Activity 2nd Group discussion : a group discussion (zoom meeting) is held with the participation of 11 students who have chosen the topic “women's issues.”
10th Week January 2, 2021	Violence against women	All	10th Activity 3rd Group discussion : a group discussion (zoom meeting) is held with the participation of 8 students who have chosen the topic “violence against women.”

was taught during the academic year prior to the main application (April 11–September 12, 2019). As a part of this process, students' critical thinking skills were determined in the context of social problems. Students were asked to complete forms prepared by the researcher in four different courses: "Population Problems," "Refugee Problems," "Women's Problems," and "Women and Education Issues." A rubric created by the researcher based on 35 critical thinking skills suggested by (Paul et al. 1989) was used to analyze the forms descriptively. As a result of the descriptive analysis, four critical thinking skills needed to be developed, and the skills in which students often encounter difficulties were identified.

The action research was conducted during the fall semester of the 2020–2021 academic year. First, a "pre-assessment" study was developed and administered within the framework of the four critical thinking skills identified in the previous year. Content and descriptive analysis were performed to analyze the "pre-assessment" study. The students had difficulties with critical thinking, as revealed by their findings. As part of the process of eliminating identified problems and developing critical thinking skills, a "system for identifying arguments and other materials and writing pedagogy" was designed, and the necessary data collection tools were prepared. Since the university where this study was conducted decided to offer distance education, educational activities and data collection tools were made compatible.

3.1.2. Research environment

3.1.2.1. The contemporary world problems course. This research was carried out in the Contemporary World Problems course, taught as an undergraduate course in the Social Studies Education Department of Nevşehir Hacı Bektaş Veli University Education Faculty in the 2020/2021 academic year. Contemporary World Problems was an important undergraduate course for preservice social studies teachers, who were considered to have an active role in raising thinking and questioning citizens.

Table 2

3.2. Study participants

Among the study participants were final-grade students in the Social Studies Education Program, an academican teaching Contemporary World Problems at the undergraduate level, and the researcher.

3.2.1. Students

A purposive sampling method, "criterion sampling," was employed in this study. The following criteria were adopted:

1. Being a 4th-year student in the Social Studies Teaching Undergraduate Program in the 2020/2021 Academic Year
2. Experiencing trouble with the problem of research in the classroom environment
3. Being a volunteer to participate in the research.

In total, 30 students participated in the study, of whom 23 were female and 7 were male. By age, 28 were between the ages of 21 and 22 years, while 2 were 34 years old. There were 19 graduates from Anatolian High School, 7 from a vocational high school, 3 from Imam Hatip High School, and 1 from a private high school.

3.2.2. Academican instructing the undergraduate course

In the action research process, the academican instructing the course collaborated with the researcher to implement the action plans designed by the researcher as part of an ongoing communication network. This academican presented the course content pertaining to Contemporary World Problems through the university's distance learning platform Perculus.

3.2.3. Researcher

The researcher participated in the action research process as a "participant-observer." A participant observation strategy is defined by (Denzin 2009) as one that incorporates "document analysis, interviews with participants, direct participation, observation, and introspection" (pp. 185–186) at the same time. Throughout the research project, the researcher prepared and implemented all course contents, action plans for developing the determined critical thinking skills, and the necessary data collection tools.

3.3. Data collection tools

As part of the data collection process for the research questions addressed within the framework of the main problem of the study, student products, individual interviews, focus group interviews, Zoom courses, and WhatsApp correspondence were used.

3.3.1. Student products

In this context, five different studies were used for data collection.

3.3.1.2. "The form of identifying the argument from the other material". One of (Cottrell's 2005) strategies for developing critical thinking skills, identifying the argument and distinguishing it from other elements, was used in this study as a form of analysis. It was expected that through the use of this method of identifying arguments, critical thinking skills would be enhanced through distance

education. As part of this experiment, newspaper articles, essays, and TEDx talks were analyzed using the “**Form of Identifying the Argument from the Other Material.**” This form was employed five times during the implementation process (see [APPENDIX A](#)).

3.3.1.3. “Writing and listening form”. Giroux’s writing pedagogy was employed as a model for integrating writing and critical thinking. During the course, the students were encouraged to view information from different perspectives and dimensions, as well as to consider information to be problematic, and therefore, an object of research. This was accomplished by using the “Listening and Writing Form.” This three-stage form was used twice at the beginning of the implementation process ([APPENDIX B](#)).

3.3.1.4. Essays. As part of the general assessment conducted after the First Action Cycle, essay writing was introduced to encourage students to write an overall article instead of the “Writing and Listening Form.” Throughout the application process, the essay-writing activity was conducted twice.

3.3.1.5. “Semi-Structured group discussion form”. Three online discussion sessions were conducted as part of the Fourth Action Cycle. As part of the discussion sessions, a “Semi-Structured Group Discussion Form” was prepared and administered under the moderation of the researcher (see [APPENDIX C](#)). On the Zoom platform, discussions were held in sessions lasting approximately two hours and 20 min each.

3.3.2. Interviews

Individual interviews were conducted at three different time periods to provide feedback and evaluate students during the action research process: two at the beginning of the application process and one at the end. Accordingly, individual in-depth interviews were conducted with nine students at the beginning of the implementation process and eight students during the subsequent period. Following the conclusion of the activities in the third stage, seven students were interviewed. The interviews were held using the “Semi-Structured Interview Form” (see [APPENDIX D](#)).

3.3.3. Focus group interview

Individual and focus group interviews provide different and complementary means of obtaining information [Patton, \(2014\)](#). Four focus group interviews were conducted with 20 students at the end of the research process to provide a general assessment of the implementation. Four students participated in the first interview, six in the second, and five each in the third and fourth interviews. Under the researcher’s moderation, interviews were conducted on the Zoom platform using the “Semi-structured Focus Group Interview Form” (see [APPENDIX E](#)). The first focus group interview lasted 84 min, the second 110 min, the third 86 min, and the fourth 130 min.

3.3.4. Document analysis

According to [\(Yıldırım and Şimşek 2016\)](#), written and visual materials as well as materials related to the problem could be included in the research along with interviews and observation methods when direct observation and interviews were not possible in qualitative research or to increase the validity of the research. During the course of document review, the WhatsApp correspondence between the researcher and students throughout the process and the Zoom lessons in which the researcher addressed critical thinking skills were turned into a dataset.

3.4. Data analysis

Two-dimensional spiral, micro, and macro data analyses were followed during the action research process. As part of the microscale analysis, descriptive analyses were performed to progress to the subsequent processes during the collection of research data. During this stage, the extent to which critical thinking skills were acquired and the problems encountered were determined after each activity, and information was gathered regarding what needed to be done in the next phase. The macroscale data analysis spiral included an analysis of all data obtained after completing the action research process. By analyzing all the data together, the relationships between themes and categories were revealed, and conclusions were drawn. All data were organized into computer files based on written documents of WhatsApp correspondence and transcriptions of individual and focus group interviews.

The transcripts obtained from the data collection tools were analyzed using the MAXQDA 2020 qualitative analysis program. Computer-aided data analysis software is recommended for large data sets [Merriam, \(2015\)](#).

3.5. Validity and reliability

3.5.1. Long-Term interaction

A preliminary study was conducted in the fall semester of the academic year 2019/2020 in order to determine the critical thinking skills that should be developed. The main implementation of the research took place between October 2020 and January 2021. Therefore, it is possible to state that the data collection process took a very long time from beginning to end.

3.5.2. In-Depth oriented data collection

Because the content of this research was multidimensional and complex, including critical thinking skills, detailed and in-depth

data collection methods were used instead of standard measurement techniques. The researcher analyzed the data from a comparative and interpretive perspective using critical reflection to understand the implementation process and improve the interventions.

3.5.3. Variability

A total of two variations were used in this study, namely the “method” and “analysis” variations. Interviews, focus-group interviews, and document reviews were used for method variation. For the analysis variation, a combination of descriptive and content analysis was conducted through the use of “Graded Scoring Rubrics.”

3.5.4. Expert analysis

Expert analysis was carried out by members of the “validity committee,” which was composed of academicians working in the Education Faculty, Educational Sciences, and Primary School-Preschool Departments of Çukurova University.

Another application included in the expert review was the coding of the data by a second expert and the verification of the agreement rate between the two coders. Accordingly, an academician at Nevşehir Hacı Bektaş Veli University carrying out studies on global citizenship education and critical thinking analyzed the data obtained from five different data collection tools (student product, individual interview, focus group interview, online group discussion, WhatsApp correspondence). A total of 39 pages of the data set were independently coded by another academic line by line, including 10 pages from the “focus group interview,” 3 pages from the “student product,” 4 pages from WhatsApp correspondence, 10 pages from the “individual interviews,” and 12 pages from the “group discussions.” Both researchers reviewed the coding side by side and discussed the points on which they disagreed. Based on the number of disagreements, the agreement rate was calculated using the reliability formula proposed by Miles and Huberman (1994, p. 64).

$$\text{Reliability} = \frac{\text{number of agreements}}{\text{total number of agreements} + \text{disagreements}}$$

According to the calculations, there was a 0.90 agreement rate between the coders.

3.6. Role of the researcher

The researcher participated in the action research process as a “participant observer” Denzin, (2009). The researcher prepared and implemented the course content, action plans for developing critical thinking skills, and necessary tools for data collection. As part of the action plans, the researcher held Zoom courses and online group discussions with students and gave them feedback by continuously and comparatively analyzing student products. The researcher also conducted individual and focus group interviews with students throughout the process. All activities and practices were carried out by the students and researcher, except for PowerPoint presentations containing the subject matter presented by the academician lecturing the course.

4. Findings

The findings are presented in two stages. The first focuses on the changes in critical thinking skills, followed by the second, addressing the challenges encountered during the implementation process. In the descriptive analysis, the abbreviation (f) is used when frequency information is included. Hence, (f:8) means: 8 participants.

The first cycle.

In the first cycle, the first of two strategies, “Exploring and Implementing Critical Thinking Based on Identifying the Argument,” was implemented as the fundamental method throughout the process. The critical thinking skills discussed in this context are as follows.

1. “Analyzing or assessing the arguments, explanations, and opinions.”
2. “Developing one’s own perspective: Argument development and giving details.”

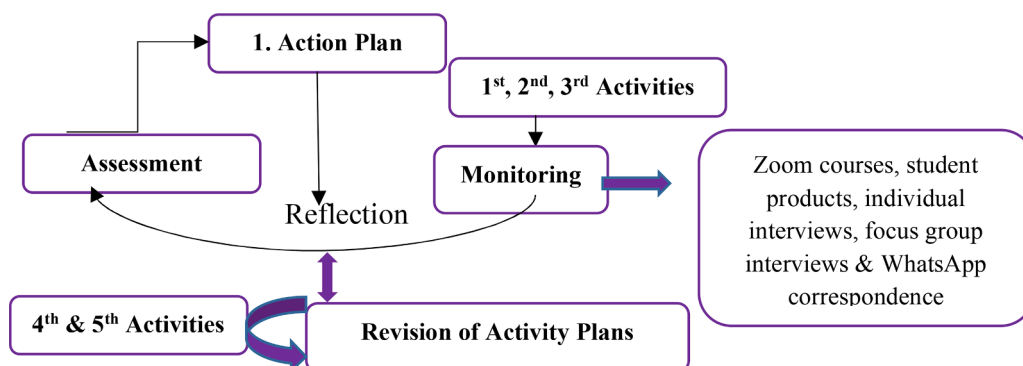


Fig. 2. The procedures followed for the development of relevant critical thinking skills in the first action plan.

The studies conducted on the development of critical thinking skills in the First Cycle are presented in Fig. 2.

4.1. The transformation in “Analyzing or evaluating arguments, explanations, and opinions” transformed and the difficulties experienced

In the action research process, the first step consisted of introducing the first two basic strategies, an approach based on distinguishing arguments. Different materials such as essays, newspaper articles, and TEDx speeches were used to practice distinguishing arguments and other items. Based on the data collected in this cycle, components such as Zoom lessons, individual in-depth interviews, focus groups, and WhatsApp correspondence were analyzed by content analysis, and student products were analyzed by descriptive analysis. Appendix F provides a detailed description of the themes that emerged from the content analysis. The extent of change in this critical thinking skill was determined according to two main themes: (1) understanding the approach to distinguishing arguments and (2) epistemological awareness as a cognitive frame of reference.

4.1.1. Understanding the approach to distinguishing arguments

In this category, the students comprehended distinguishing the argument in two distinct stages: *the triggering event* and *discovery & integration*.

4.1.1.6. Triggering event. At the beginning of the distance education applications, students experienced internal discomfort and surprise when they encountered the task of distinguishing arguments, a study method they were not familiar with and that they had not expected. Students experienced this stage by following the processes of *inquiry*, *complexity*, and *awakening* in their minds. The following thoughts conveyed by a student while searching the Internet for the content used in the applications clearly illustrate the nature of their questioning process.

“Please direct me to where you find them (the course contents) since it would be easier for me to search from that point... Other teachers usually provide information, and we find it and write it down... We used to see it in exams all the time, but I was unable to locate the information you provided, so I began taking your course.” (Individual interview, Participant 10 [P10])

4.1.1.7. Discovery & integration. Astonished by the manner in which the arguments were distinguished in the first stage, the students attempted to comprehend the strategy at this stage. They began to use an approach to distinguish arguments after ignoring the complexity of the early stages. By creating a creative synthesis of the acquired knowledge, the students developed a method for integrating their knowledge. Students encountered this category by following the stages of *slowly comprehending the approach*, *exploring the approach*, and *overcoming the challenge*.

“I think I’ve improved now; I can more easily find (arguments). I had difficulty identifying the argument in my previous attempts. I saw you explaining the answer key later when I entered your class. Here, we had an argument, stereotypes, and discrimination statements. As soon as I noticed them, I paid attention to my other writing.” (Focus group interview, P12)

4.1.1.8. Functionality of the feedback. Based on the feedback they received from the researcher, the students realized the mistakes they had made and understood the points they needed to pay attention to in the future. During the feedback process, students who realized their errors stated that they would like to conduct the related study again.

“I really appreciated the opportunity to meet with you one-on-one, and I look forward to reflecting on the exam as a result. At least, I was able to recognize my own mistakes. My own mistakes allowed me to gain a deeper understanding of the subject.” (Individual interview, P24).

4.1.2. Epistemological awareness as a cognitive frame of reference

The categories within the scope of this theme indicate that the students comprehended distinguishing the argument and used the system in other cognitive tasks. The transfer problem was one of the greatest challenges in learning critical thinking skills. The ability to transfer skills from one situation to another is known as transferring skills. Students developed epistemological awareness when they applied distinguishing the arguments in newspapers or essays in their everyday lives.

“In general, I’m interested in trending news, and I read a lot. I try to identify arguments if there are any. This is something I have observed in myself. In an effort to understand the thoughts of the author, I am attempting to identify them directly.” (Focus group interview, P14).

4.1.3. The difficulties encountered

Content analysis of online learning environments revealed two major themes regarding students’ difficulties and areas for improvement: (1) silence and (2) complexity. As a result of the descriptive analysis of student products, it was determined that students had difficulty meeting the criteria of “finding whether different perspectives were addressed” and “identifying the argument.” The number of students who answered the “identify the argument” criterion incorrectly was eight (f:8). Upon reviewing the answers of the students who answered the argument-finding criterion incorrectly, it was discovered that these students selected explanation-inference expressions that the speaker (TEDx talk) indirectly addressed and used to strengthen the main argument rather than identifying the main argument themselves. In fact, the frequency of errors in the criterion “identifying whether different perspectives were included” (f:11) supported this conclusion. Students who were unable to identify the speaker’s point of view were unable to identify the thesis put forward and determine whether different perspectives were presented in the article.

4.2. Transformation and experienced difficulties in the skill of “Developing one’s own perspective: argument developing and giving details”

The students analyzed the arguments and other items using essays, newspaper articles, or TEDx talks. On the other hand, students wrote factual statements about the world problem discussed, and wrote explanatory inference sentences using information from multiple sources. While the students were dividing a whole formed by the bringing together of various elements, on the other hand, they were engaged in the process of writing in order to reestablish a holistic structure on their own. Two main themes determined the change in this critical thinking skill: (1) metacognitive thinking: self-assessment of writing, and (2) metacognitive thinking: self-assessment of perspectives. For the categories and subcategories of the identified themes, see [Appendix F.1](#).

4.2.1. Metacognitive thinking: self-assessment of writing

During the process of distinguishing the argument, the students gained a greater understanding of the author’s strategy for discussing the argument, as well as how the argument was structured. They were then given the opportunity to write their own writing by assuming the same role as the author and providing a similar perspective. Students began employing both strategies in a dynamic relationship network.

“Previously, we wrote and delivered compositions, but not in this way, giving no details. As an example, we did not pay attention to the argument, we did not include a different perspective, and we did not draw any inferences ... Before, we simply wrote whatever was in our minds, adding an introduction, development, and conclusion... It was during this term, however, that we learned to write with arguments, then we learned to add additional information, and we learned to base our arguments on evidence ... Before, we were just writing down what we thought.” (Focus group interview, P25)

4.2.2. Metacognitive thinking: self-assessment of perspectives

The process, which started by distinguishing the argument and other items, continued with the students first writing factual statements, explanations, and inferences, and then gaining argument-based writing skills by adding their original thesis. By interacting with the different perspectives of the sources discussed, the students, who researched world problems, were able to incorporate factual statements.

“It seems that I have gained a broader perspective on a number of issues... Specifically, on today’s global issues, for example, a problem we are experiencing right now. If I were looking at an area of 20 m²s related to today’s world problems, now I am looking at an area of 50 m²s after your lecture.” (Individual interview, P2).

4.2.3. Difficulties encountered

As a result of the content analysis, the following themes were identified as the areas where students had difficulty and needed improvement with regard to this critical thinking skill: (1) non-comprehensive comprehension, (2) inability to develop a personal perspective, and (3) non-comprehensive understanding of knowledge and the categories relating to these themes. Although the students learned how to use the two basic strategies of the action research process, they encountered difficulties in proposing an original thesis. This was because they believed that the explanation-inference sentences they discussed were responsible for their arguments. To overcome the difficulties encountered in writing and to improve students’ original theses through a more flexible writing approach, studies based on composition and reflection were emphasized at this stage of action research.

The second cycle. As part of the Second Action Cycle, the action was taken for “Developing Comprehensive Insight: Integrating Argument Distinguishing & Writing Pedagogy.”

The critical thinking skills discussed in this context were as follows:

1. “Questioning the Reliability of Information Source,”
2. “Avoiding Simplifications and Specifying the Generalizations.”

Fig. 3 illustrates the processes involved in the development of critical thinking skills during the Second Cycle of Action.

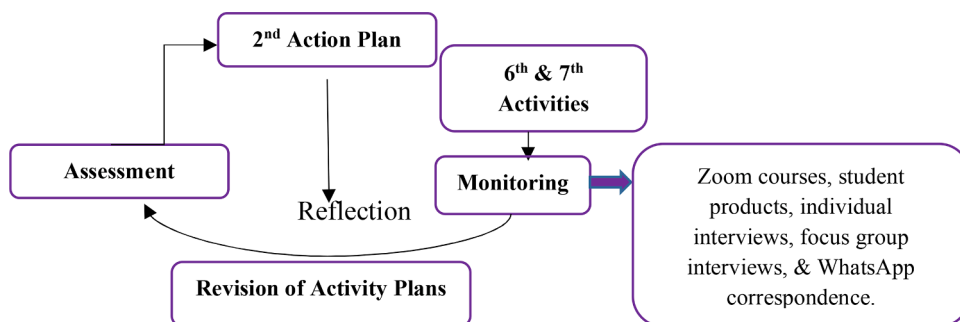


Fig. 3. The processes that were conducted on the development of critical thinking skills during the second cycle of action.

4.3. Transformation and difficulties experienced in the skill of “Questioning the reliability of the information source”

The evolution of this critical thinking skill can be characterized by two main themes: (1) developing a critical view of knowledge and (2) utilizing rationalist epistemology. For detailed information on the categories and subcategories of the themes, see [Appendix F.2](#).

4.3.1. Developing a critical view of knowledge

It is critical to note that the frequent use of distinguishing the argument in the activity process led students to question the evidence on which the argument was based. Students who gained skills in analyzing and assessing arguments reported that they developed reliable questioning procedures and developed a critical viewpoint when approaching the information. The results of our study showed that students felt compelled to examine information from various angles by assessing its depth and breadth from different perspectives.

“There are certain aspects of the news that we do not pay attention to. Whenever I see a news article, I pay attention to the date, to see if it has been updated. As I read the news, I used to look at the date but did not pay attention to whether it had been updated or not. Now that I am looking at it directly, I would like to know whether it has been updated recently.” (Focus group interview, P6).

4.3.2. Utilizing rationalist epistemology

The understanding of rationality upon which critical thinking is based, as well as strict adherence to scientific evidence, may reveal an approach that is intended to leave various “ways of knowledge” in the background. According to the categories within this theme, students placed a high value on objectivity and the need for scientific evidence, with an emphasis on the positivist tradition. Based on the findings of this study, the students did not believe that validation forms such as experience and emotion were valid and did not incorporate these methods into their research or writing.

“For the information source to be considered reliable, it should provide numerical data or use a similar expression. As long as it is backed up by a really strong article, I believe I am similarly inclined in real life as well. Nevertheless, if it comes up with an idea but does not fill the ground, I do not trust it. This is true regardless of the subject.” (Focus group interview, P24)

4.3.3. Difficulties encountered

Based on descriptive findings related to the “Questioning the Reliability of the Information Source” skill, it can be concluded that students were unable to meet some criteria related to impartiality, and in this context they were not able to identify the point of view from which the news was derived. In the course of the content analysis, it was found that students made assumptions in a positivist manner in their writing while presenting personal observations as factual information. As a result of a discriminatory approach based on nationalist ideology, some students experienced critical thinking barriers.

4.4. The transformation and difficulties experienced in “Avoiding simplifications and specifying generalizations”

The transformation in this critical thinking skill was observed within the scope of two main themes: (1) in-depth understanding and (2) perspective development on social reality. For detailed information on the categories and subcategories of the themes, see [Appendix F.3](#).

4.4.1. In-Depth understanding

As part of the writing pedagogy, students were required to incorporate factual expressions in their writing as well as explanation-inference sentences within the scope of the factual expressions they addressed. To be able to write factual statements, the students conducted research using a variety of sources related to the topic. Through this process, the researcher established individual connections with students who needed feedback, and after helping them realize their mistakes, she tried to control the process by assigning revised writing assignments. Students were able to develop their intellectual capacity through this research-process-based writing process. According to the categories in this theme, the students adopted a research-oriented approach, avoided self-centered explanations, and approached the problem from multiple perspectives.

“Before, I would write whatever came to my mind. Now I do a little more research, I filter the information a little more, I put it in my mind, and I write it. I conduct research, I read, and I begin to provide examples.” (Focus group interview, P14)

4.4.2. Perspective development on social reality

In accordance with the categories determined within the framework of this theme, students developed arguments emphasizing the relationship between social pressure structures and relations of dominance, as well as criticizing state policies. Students who were critical of the conservative structure and hierarchical nature of information interactions in society demonstrated that they had acquired an understanding of social reality. During the action research process, students were required to acquire factual information regarding the world problem, as well as to define the problem by identifying the causes along multiple dimensions. As a result of the argumentation-based approach, the students were able to develop reliable inquiry procedures related to global issues, and to develop perspectives on social reality because of these factors.

“Teacher, when we look at the world problems of today, we can see that we are under state pressure as citizens. It is the governments that are responsible for creating world problems, not us. Therefore, our role is merely one of an intermediary.” (Individual interview, P17)

4.4.3. Difficulties encountered

Regarding this critical thinking skill, students “presented personal observations as factual knowledge” within the context of critical thinking barriers. Several students who had difficulty implementing writing pedagogy made ungrounded statements, used ambiguous expressions, and attempted to write with superficial information without giving place to introductions or background information.

The Third Cycle

The third action cycle was undertaken with the objective of “**Adapting the Argument Distinguishing & Writing Pedagogy of Listening and Speaking.**” This was accomplished through three online discussions. The objective of the discussion sessions was to assist the students in proposing original theses and structuring these theses by providing the necessary evidence, explanations, and different perspectives. In other words, students were intended to adapt the critical thinking strategies they used throughout the speaking activity.

Fig. 4

4.5. Transformation of critical thinking skills and challenges experienced in online discussions

Two themes were identified in the online discussions: (1) developing a personal perspective and (2) structuring the thesis. For detailed information on the categories and subcategories of the themes, see [Appendix F.4](#). It was evident from the discussion sessions that students developed a distinctive point of view using alternative methods of knowledge, such as emotion and experience. Here is what one student shared about his grandfather’s experience with Syrian refugees.

“Based on what I have learned from my uncles, we came to this country during the War of Independence. The atmosphere was quiet here... In a discriminatory context, they were also fleeing the war. My grandfather had daughters, but no sons. Out of necessity, he brought four daughters and his wife to this country. Our approach should be based on empathy. We need to take into account the obligations (toward Syrian refugees).” (Individual interview, P2).

4.5.1. Developing a personal perspective

Students gained new perspectives, identified different perspectives, and developed original thesis statements within the context of this theme.

“One of our friends commented that as education increases, violence decreases, and women’s rights are more widely recognized. According to a study conducted by Kadir Has University, egalitarian attitudes do not increase with education. Compared with 80 women murdered in 2008, 474 women were killed in 2019. In other words, the level of education did not, however, result in a decrease in violence and death. Why?” (Focus group interview, P19)

4.5.2. Structuring the thesis

Based on the categories within the scope of this theme, students included facts and explanations related to the subject as well as inferences and explanations.

“Based on the official data, I observe that the majority of employees are male. The official data for 2020 indicates that while there are 30,633,000 people employed, only 9 million are women. 20 million men. Women continue to be dependent on men for their livelihood. It is because they believe they will be rejected by society...” (Focus group interview, K16).

4.5.3. Difficulties encountered

In the first discussion session, we found that some students had difficulty developing a personal perspective, understanding the reasons for and details of the topic, and presenting an original thesis. In light of the fact that Syrian refugees were the subject of discussion, it was determined that some students took an approach based on prejudice and discrimination. During the online discussion, there were interactions such as making assessments from a nationalist viewpoint (f:4), expressing opinions based on prejudice

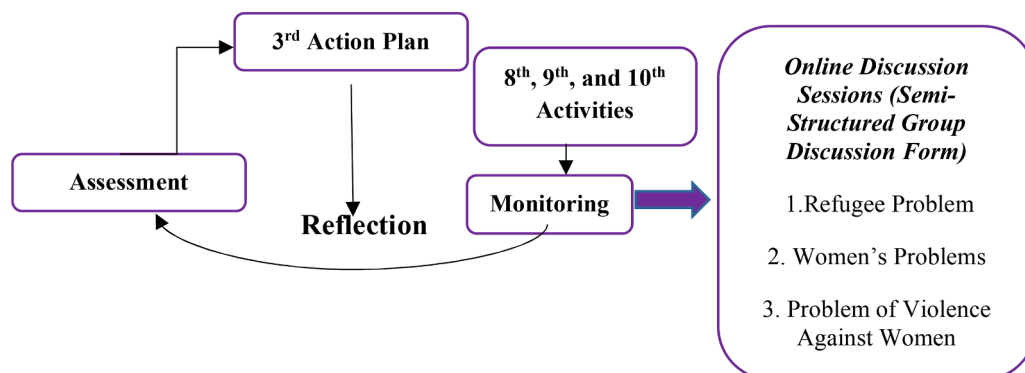


Fig. 4. The processes that were conducted for the development of critical thinking skills during the third cycle of action.

(f:2), and expressing opinions based on gender discrimination (f:1).

Based on the second discussion, it was determined that the difficulties identified in the first session were reduced to a certain extent but continued in this session. It was observed that students had difficulty developing personal perspectives, presenting theses, developing perspectives, and listening critically. According to the findings, some students had cognitive disabilities (f:6) and positivist perspectives regarding information sources (f:2).

Students experienced significantly fewer difficulties in presenting their theses in the third discussion session than the previous ones. There was evidence that some students had difficulty presenting their theses, were unable to structure them (f:4), lacked theoretical knowledge of the subject (f:4), and were unable to listen critically (f:3).

5. Discussion

Through critical thinking, citizens can make critical and conscious contributions to their society. The cognitive ability to analyze and discuss social problems is essential for preparing students to achieve this goal. (Larson et al. 2009) assert that developing students' ability to read and assess the arguments of others is a vital civic skill in a democracy. For a democracy to function effectively, citizens must be able to reason and be knowledgeable about public issues.

In this context, when the findings obtained in the first action cycle were examined, it became apparent that the students experienced uncertainty and complexity in distinguishing the argument, as well as difficulty determining whether different perspectives were included in the studies discussed. The difficulties experienced by the students in identifying arguments were consistent with the findings of (Arum and Roksa's 2011) study, which included more than 2300 students; at least 45 percent of the university students in this study failed to acquire statistically significant critical thinking, complex reasoning, or writing skills according to the study. The results of another study indicated that a large number of university students did not acquire the ability to understand, assess, and write arguments before leaving university Larson et al., (2009).

The fact that students did not work on the items in question could be attributed to their inability to identify arguments. Larson et al.'s (2004) study, which involved 57 native English speakers, concluded that university students without training in general argumentation skills failed to identify structurally flawed arguments and acceptable arguments without training in general argumentation skills Larson et al., (2004, p. 358). The study conducted by Larson et al. (2009, p. 358) in three experimental studies determined that university and high school students were unable to assess arguments in terms of quality. However, a short-term special training program improved students' ability to identify failed arguments, but immediately following the training, students still required feedback. In fact, the category of *function of feedback*, which was among the findings of this action research, revealed that students' individual and group feedback played an active role in identifying arguments. Students were provided with feedback by the researcher during the action research process, which enabled them to realize their mistakes and gain a clear understanding of how the system was used.

Although students stressed the importance of feedback as part of the action research process, (Van Gelder 2005) emphasized that it was difficult to gain expertise in thinking skills and that it was necessary to develop critical thinking skills. Additionally, it has been suggested that another reason students had difficulty identifying an argument during the initial stages of action research was that the actions performed required difficult operations due to their nature and that they required practice to develop these skills.

Citizens should be able to determine whether there is evidence in the arguments put forward or whether the evidence presented guarantees the outcome. Assessing the structural quality of arguments is also an important skill for students to understand others' arguments and produce their own Larson et al., (2009). In the action research process, students identified the arguments in essays, TEDx talks, and other materials, and began to understand the authors' strategies while discussing the argument and the way in which the argument was structured, and thereby acquired a similar perspective. In other words, the students began to assume the same role as the author, resulting in the creation of a new cognitive framework. As (Van Gelder 2005) pointed out, one of the greatest obstacles to learning new skills, especially critical thinking skills, is the transfer problem. The transfer of skills refers to the application of an acquired insight or skill to a new situation. Likewise, in this action research, students applied the distinguishing the argument skill to what they read and wrote every day and used the same approach for creating cognitive awareness during speaking and listening, which resulted in students transferring their critical thinking skills. Developing perspectives on social problems is recommended in the context of citizenship education and contributes to the development of critical thinking Berdahl et al., (2020); Çavdar and Doe, (2012). Writing is essential to critical thinking. In the words of (Murray 1984, p. 4), "writing is the most disciplined way of thinking. It allows us to be imprecise, to step back and clarify what we are thinking, and to identify what our words really mean." The action research process revealed that students made self-assessments of writing actions as part of their metacognitive thinking processes (see Appendix F.1). Metacognition is a cognitive competency directly related to critical thinking (Kuhn, 1999; Lau, 2015; Magno, 2010), and it involves individuals recording their thoughts Barnet and Bedau, (2011). Accordingly, the students' ability to reflect on their writing progress and the strategies they followed during the writing process indicated that they had developed metacognitive competencies. Furthermore, they expressed pleasure in writing in addition to the difficulties they encountered while writing. They approached today's global problems from a number of perspectives, including critical, human rights-based, and scientific, and developed a perspective that was interpretive in nature. As a result of the metacognitive learning processes that the students engaged in while writing, their critical thinking skills were enhanced. The students developed not only metacognitive skills in their writing but also a critical perspective on knowledge. Based on the findings related to "Questioning the Reliability of Information Sources," students began to recognize the need to examine information from multiple perspectives by addressing the depth and breadth of information and developing reliable inquiry procedures (see Appendix F.2). Students' development of reliable inquiry procedures for information is considered an important component of active democratic citizenship in light of recent attempts at online political manipulation using

targeted political propaganda and fake news [Piedade et al., \(2020\)](#).

The ability to think independently and autonomously is considered the most essential skill for critical citizens [Johnson and Morris, \(2010\)](#); [Siegel, \(1988\)](#); [Veugelers, \(2007\)](#). By controlling their cognitive processes, students develop a critical perspective toward the information they obtain from their environment, resulting in the development of their civic skills, which enables them to think independently and autonomously. [\(Kuhn 1999\)](#) defined metacognitive skills as those that require a person to have an understanding of their own and others' knowledge. Critical thinking requires the development of metacognitive skills because by definition it involves reflecting on what is known and how that knowledge can be justified. It is important for people to gain control over their thoughts to take control of their personal and social lives [Kuhn, \(1999\)](#).

[Johnson and Morris's \(2010\)](#) value of "deep thinking" and the ability to "examine causality more deeply" within the framework of critical citizenship were in agreement with our findings. When the findings related to the critical thinking skill of "Avoiding Simplifications and Specifying Generalizations," which were discussed in the action research process, were analyzed, it was observed that the students developed a deeper understanding of their writings and avoided simple self-centered explanations, and often researched and discussed the problem from multiple perspectives. (see [Appendix F.3](#)). The students attempted to examine causality more deeply in their comments by providing specific details and explanations of the problem. Additionally, our findings are consistent with those found in the literature based on empirical research. For example, [Tsui's \(2002\)](#) study found that effective pedagogies that develop critical thinking, writing exercises, and rewriting activities after receiving feedback encouraged students to think deeply, which contributed to the development of critical thinking skills.

According to [\(Brookfield 1987\)](#), critical thinking skills can be acquired through a series of steps that could be described and shared. It was possible to mention that the components that created these stages were quite similar, even though they were expressed with different concepts in the literature [Garrison, \(2017\)](#); [Garrison et al., \(2001\)](#). This action research was compatible with the literature in that the three action cycles were organized in a pattern that followed the stages of "discovery-practice," "integration," and "adaptation of the approach to what they listen to and speak." [\(Brookfield 1987\)](#) stated that the "triggering event" stage represented the first step in the process of developing critical thinking skills. At this stage, an unexpected event causes feelings of internal discomfort and bewilderment. In the first action cycle, students experienced a sequential process in the triggering event category. The students stated that they encountered an unexpected event related to the approach of distinguishing the argument with an interrogative approach and experienced complexity, and that this complexity led to a gradual mental awakening.

The "discovery" stage was described as the process of weighing options and seeking meaning, as well as being open to new information and discovering it [Brookfield, \(1987\)](#). Taking our findings into account, the students indicated that in the discovery and integration category, they slowly acquired a grasp of the approach related to argumentation-writing exercises and that they acquired an understanding of the approach by overcoming their initial difficulties. [\(Brookfield 1987\)](#) suggested that the next step was to develop "alternative perspectives." Alternative perspectives are discovered at this stage, leading to the emergence of new ways of thinking. Based on our findings, students developed critical perspectives on current world problems from both the scientific and human rights perspectives.

As part of the "integrating" stage, a variety of cognitively acquired information was synthesized in order to arrive at a new solution or make a change within the self [Brookfield, \(1987\)](#). Based on our findings, students enhanced their cognitive awareness levels through the development of epistemological awareness and the adaptation of an argument discrimination system based on what they read, wrote, spoke, and listened to. The progress of the action cycles in the study process in a pattern following certain stages is consistent with the literature.

6. Conclusion

Analysis of the research findings revealed that the use of argument discrimination writing pedagogy in a holistic approach to distance education practices, which was discussed in three phases, improved students' metacognitive competency and critical thinking skills.

Based on the transformations of the students' critical thinking skills, the conclusion was reached that students comprehended to distinguish the argument in a process consisting of "triggering events" and "discovery and integration" stages, and developed an epistemological awareness by applying this approach to what they read, wrote, spoke, and listened to as a cognitive frame of reference. Feedback was found to be quite functional during this process. Students made self-assessments related to developing their writing skills and acquiring perspectives within the context of metacognitive thinking. In conclusion, students developed an in-depth understanding of the skills related to conducting research on writing and addressing multiple dimensions of a problem.

In online learning environments, the participants initially experienced "silence" and "complexity" regarding the skill of "analyzing or evaluating arguments, explanations, and opinions," and it was difficult for them to determine whether different perspectives were included in the given material and identifying the argument of the given material. In the beginning, they had a limited understanding of the skill "Developing their own perspective: developing and elaborating arguments," they could not develop a personal perspective, and their knowledge understanding was incomplete. It was determined that they could not meet the criteria of "impartiality" and "accuracy" pertaining to the skill of "questioning the reliability of the information source" at the beginning. During the writing process, they made assumptions and were unable to assess the reliability of the source effectively. According to the results of the study, some students were found to have taken some actions that prevented them from using critical thinking skills regarding the skill of "Avoiding Simplifications and Specifying Generalizations." These students provided unsubstantiated explanations and used ambiguous expressions in their writings to convey superficial information. Based on the findings of the last action cycle, it was determined that they had difficulty in developing a personal perspective and suffered from cognitive impairment.

7. Recommendations

By addressing current world problems using critical thinking skills, students analyzed essays, TEDx speeches, and newspaper articles on certain topics; contrariwise, writing, which begins with research, provided students with an opportunity to develop meta-cognitive competencies and identify critical thinking skills. To this end, the findings suggest that students' critical thinking skills and citizenship competencies should be developed through a similar approach to issues related to global problems in higher education programs.

The findings further suggest that argument-distinguishing and writing pedagogy can be usefully integrated into the distance education processes of the faculty of education. Moreover, feedback-oriented online courses and semi-structured online discussion sessions should be conducted to enable students to adapt to the cognitive framework they have acquired in their listening and speaking.

CRedit authorship contribution statement

Arzu Saldıray: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Ahmet Doğanay:** Writing – review & editing, Supervision, Methodology.

Data availability

Data will be made available on request.

Acknowledgments

This study was funded by Çukurova University Scientific Research Projects Coordination Unit as a Ph.D. thesis project with the code SDK-2021–13649.

Appendix A The Form of Identifying the Argument and Other Material

Appendix A

The form of identifying the argument and other material

1. WHAT IS THE AUTHOR'S ARGUMENT (THE PROPOSED THESIS)?

2. What evidence and reasons did the author give to support the argument (The proposed thesis)?

3. Are different perspectives included in the article? What are these perspectives?

4. Please provide information about the "Introduction" part of the article.

5. Are definitions included in the article? If so, what are these definitions?

6. Give two examples of the explanations and inferences it discusses.

7. Did the argument give summary information? What was this summary? Please answer with an example.

8. Is there any background information in the article? If yes, please give brief information.

Graded Scoring Rubric for "the form of identifying the argument and other material"

criteria	Score
1. Identifying the argument	15
2. Identifying evidence and reasons to support the argument (the proposed thesis)	15
3. Identifying if different perspectives are included	15
4. Identifying the introduction	11
5. Identifying the definitions	11
6. Identifying explanations & inferences	11
7. Identifying the summary	11
8. Identifying the background information	11

Appendix B Listening and Writing Form

Listening and Writing Form

1. Listen to Larry Brilliant's TEDx talk on "The Case for Optimism." What are the 5 important facts that caught your attention in his talk?

2. Write a two-paragraph essay integrating 5 new points you have acquired.

3. Write an essay expressing your thoughts (opinions) within the scope of the paragraphs you write above?

Appendix B 1

Graded scoring rubric for the “Listening and writing form” criteria	
Score	
1. Distinguishing fact from opinion and stating it in a meaningful sentence	5 × 2 = 10
2. Identifying and summarizing the problem/issue related to the topic	10
3. Making explanations and inferences based on facts	10
4. Explaining the main reasons and key concepts clearly	10
5. Using critical vocabulary	10
6. Clearly reflecting one’s point of view on the subject	10
7. Clearly addressing the reasons for the point of view	10
8. Clearly stating the conclusion (argument) statement and basing it on evidence	10
9. Incorporating different perspectives	10
10. Associating the topics discussed with each other and the result	10

Appendix C. Semi-structured group discussion form

Discussion no.	Discussion questions
1st Discussion Session Refugee Problem	1st Cycle: <i>Voice Cycle 1</i>: If I asked you to describe your perspective on Turkey’s recent refugee movement, what would you say? (Thinking time: 3 min) In depth: In what ways did refugee mobility affect Turkey positively and negatively? 2nd Cycle: <i>Voice Cycle 2</i>: Choose one of your friends who spoke in the first cycle, “Are there any arguments in what that person says? What is the proof? Are you able to catch this speaker’s point of view? In your opinion, what can be the reasons behind this viewpoint?” (The speaker will not mention his/her own views). 3rd Cycle: <i>Answering Cycle 1</i>: The question is posed: “What can you say about refugee mobility from the perspective of refugees and locals, the international community, and the countries they migrate to?” Following the first speaker, the second speaker expresses his/her opinions, which build on the previous speaker’s remarks. Speakers are free to agree or disagree with previous speakers’ views. In depth: What were the consequences of refugee mobility for local communities? For refugees, what were the consequences of their arrival in our country? 4th Cycle: <i>Answering Cycle 2: Open Discussion</i>: Questions to be asked for each participant to answer: “What are your thoughts about why the person who spoke before you view the subject in this way?” Do the speeches give you a new perspective/explanation? Could you tell me about the point that is most challenging for you?

Appendix D. Semi-structured interview form

1. What steps are you taking to identify the argument in the given material/paragraph/speech? Do you have a strategy in place?
2. Have you identified evidence that supports the argument? Have you encountered any difficulties?
3. Would you be able to check whether the author or speaker has a different point of view? What is your assessment of yourself in this regard?
4. Can you tell me if there is any point that you do not understand or have difficulty distinguishing the argument from other items (other material)? Are there any areas where you feel you need to improve?
5. Is there a possibility that you may be able to discuss the subject from various perspectives in your essays? Is there anything you have observed about it? What new points have you noticed?
6. Are you confident that you will be able to develop your own perspective? In the course of the process, what achievements have you made so far?
7. Can you identify any points where you feel you are not developing your own perspective effectively? Could you please provide more information about them if so?
8. Can you discuss the similarities and differences between the studies you have conducted in the past and the studies that have developed critical thinking that we are implementing today?
9. Could you please share with us your thoughts regarding the impression that our efforts to develop critical thinking skills have left on you? What is your assessment of the process from a positive or negative standpoint?

Appendix E. Semi-structured focus group interview form

1. How well do the skills we studied in the Today's world issues course on critical thinking match your expectations of this course?
2. Have the assignments you completed during the course (Essay analysis, TEDx talk analysis, analysis and comparison of newspaper news, essay writing):
 - a. enabled you to analyze the argument?
 - b. enabled you to develop your own perspective?
 - c. enabled you to question the reliability of the information source?
 - d. enabled you to avoid simplification and specifying generalizations?
3. Are you familiar with our studies from previous courses? Are there any similarities or differences between them?
4. Have you encountered any difficulties while conducting the studies? If so, what are these difficulties? What might be the cause of these difficulties?
In depth: Capturing different perspectives? Being multidimensional in texts? Including details in texts?
5. According to your definition of democratic citizenship, how do you think the studies we have conducted in connection with this process relate to these characteristics?

Appendix F. Content analysis findings related to the skill of “analyzing or evaluating arguments, explanations and opinions”

Themes	Categories	f (frequency)
Comprehending the Argument distinguishing Approach	Triggering Event	Inquiry process
		Complexity
		Awakening in the mind
	Discovery & Integration	Slow adoption the approach
		Exploring the approach
		Overcoming the challenge.
Functionality of the Feedback	Recognizing the mistake and speaking up	
	Emphasis on the explanatory function of feedback	
	Feeling the desire to prepare new work with feedback	
Epistemological Awareness as a cognitive Frame of reference	Applying the argument distinguishing approach to what they read/write	
	Applying the argument distinguishing approach to what they speak/listen	
	Argument distinguishing approach creates a cognitive awareness	
	Developing a perspective that increases awareness	

Appendix F.1. Content analysis findings related to the skill of “developing one’s own perspective: argument developing and giving details”

Themes	Categories	Subcategories	f
Metacognitive thinking: Self-assessment of writing	Reflection on the progress he has made in his writing	Realizing progress in your writing	30
		Reflecting on the error/deficiency in his/her writing	17
		Feeling pleasure and enjoyment from writing	11
		The aesthetic power of writing	9
		Reflecting on how he/she followed a strategy while writing	5
		The power of flexibility: The effectiveness of release in writing	3
		Reflecting on the purpose and approach of his/her writings	2
Metacognitive thinking: Self-assessment of Perspectives	Acquiring different frames of Reference	Developing a critical perspective on contemporary world problems	15
		Developing a perspective based on human rights	15
		Developing an interpretive perspective	6
		Including different perspectives in their writings	5
		Developing a scientific perspective	3

Appendix F.2. Content analysis findings related to the skill of “questioning the reliability of the information source”

Themes	Categories	f
Developing a critical view of knowledge	Developing trusted interrogation procedures	19

(continued on next page)

(continued)

Themes	Categories	f
Utilizing rationalist epistemology	Acquiring a problematic approach to knowledge as an object of research	16
	Recognizing ideological distortions	12
	Addressing knowledge in depth and breadth	11
	Feeling the need to look at information from different perspectives	4
	The positivist tradition: Evidence must be scientific	12
	Positivist emphasis on objectivity	6
	Reject forms of validation such as experience/feeling	1

Appendix F.3. Content analysis findings related to the skill of “avoiding simplifications and specifying generalizations”

Themes	Categories	Subcategories	f
In-depth understanding	Avoiding simple self-centered explanations	Integrate links to personal experience to provide evidence in their comments	18
		Emphasizing emotion and experience as alternative sources of information	4
		Spontaneous need to conduct research	20
		Engaging in research to structure the thesis	8
	Conducting research	Addressing issues with multiple dimensions in their comments	8
		Include details and reasons for problems in their comments	8
		Being aware of the structures of oppression and relations of domination in society	8
		Criticizing government policies	4
Perspective development on social reality	Awareness of state hegemony	Criticizing the conservative structure	6
		Questioning hierarchical information interaction	3
	Questioning dominant social norms		

Appendix F.4. Content analysis findings on how critical thinking skills are transformed in online discussions

Themes	Categories	Subcategories	f
1 st , 2 nd , and 3 rd Online Discussion	Developing personal perspective	Structuring the thesis	66
		Making an original thesis from a different perspective	
	Structuring the proposed thesis	Making explanations & conclusions about the topic	75
		Explaining relevant facts and reasons	
	Metacognitive thinking practices	Acquiring new frames of reference	19
		Metacognitive thinking practices	

References

- Acun, İ., Demir, M., & Göz, N. L. (2010). Öğretmen adaylarının yurttaşlık yeterlilikleri ile eleştirel düşünme becerileri arasındaki ilişki. *Journal of Social Studies Education Research*, 1(1), 107–123.
- Altuntaş, İ. N. (2016). *Sosyal Bilgiler öğretmen adaylarının Aktif Vatandaşlık kazanımları: Eylem araştırması*. Gazi Üniversitesi [Unpublished doctoral dissertation].
- Alvarez-Huerta, P., Muela, A., & Larrea, I. (2022). Disposition toward critical thinking and creative confidence beliefs in higher education students: the mediating role of openness to diversity and challenge. *Thinking Skills and Creativity*, 43, Article 101003. <https://doi.org/10.1016/j.tsc.2022.101003>
- Andrews, R. (2015). Critical thinking and/or argumentation in higher education. In M. Davies, & R. Barnett (Eds.), *The Palgrave handbook of Critical Thinking in Higher Education* (pp. 49–62). Palgrave Macmillan.
- Apple, M. W. (2006). *Eğitim Ve İktidar (Çev. E. Bulut)*. Kalkedon Yayıncılık.
- Arum, R., & Roksa, J. (2011). *Academically adrift: Limited learning on college campuses*. University of Chicago Press.
- Awan, A. S., Perveen, M., & Abiodullah, M. (2018). Critical thinking for citizenship education in the curriculum at secondary level. *Bulletin of Education and Research*, 40(1), 141–153.
- Barnet, S., & Bedau, H. (2011). *Critical thinking, reading, and writing: A brief Guide to Argument*. Bedford/St. Martin's.
- Berdahl, L., Hoessler, C., Mulhall, S., & Matheson, K. (2020). Teaching critical thinking in political science: a case study. *Journal of Political Science Education*, 17(Sup. 1), 910–925. <https://doi.org/10.1080/15512169.2020.1744158>
- Berg, B. L. (2001). *Qualitative Research Methods For the Social Sciences* (4th ed.). Allyn and Bacon.
- Bezanilla, M. J., Fernandez-Nogueira, D., Poblete, M., & Galindo-Domínguez, H. (2019). Methodologies for teaching-learning critical thinking in higher education: the teacher's view. *Thinking Skills and Creativity*, 33, Article 100584. <https://doi.org/10.1016/j.tsc.2019.100584>
- Brookfield, S. D. (1987). *Developing Critical thinkers: Challenging adults to Explore Alternative Ways of Thinking and Acting*. Open University Press.
- Brookfield, S. D. (2012). *Teaching For Critical thinking: Tools and Techniques to Help Students Question Their Assumptions*. Wiley.
- Burbules, N. C., & Berk, R. (1999). Critical thinking and critical pedagogy: relations, differences, and limits. In T. S. Popkewitz, & L. Fendler (Eds.), *Critical Theories in education: Changing Terrains of Knowledge and Politics* (pp. 45–65). Routledge.
- Castro, A. J. (2009). *Promoting Critical Multicultural citizenship: A case Study of Preparing Social Studies Teacher*. The University of Texas [Unpublished doctoral dissertation].

- Cottrell, S. (2005). *Critical Thinking skills: Developing effective Analysis and Argument*. Palgrave Macmillan.
- Çavdar, G., & Doe, S. (2012). Learning through writing: teaching critical thinking skills in writing assignments. *PS: Political Science and Politics*, 45, 298–306. <https://doi.org/10.1017/S1049096511002137>
- Çayır, K. (2014). “Biz” Kimiz? Ders kitaplarında kimlik, yurttaşlık, haklar. In Y. H. G. Kayacan (Ed.), *Ders Kitaplarında İnsan Hakları III Projesi Tarama Sonuçları*. Tarih Vakfı Yay.
- Darmanin, M., & European Students' Union (2020). *Democratic citizenship education: guidelines and recommendations for implementation in higher education*. <https://files.eric.ed.gov/fulltext/ED614458.pdf>.
- Denzin, N. K. (2009). *The Research act: A theoretical Introduction to Sociological Methods*. Routledge.
- Garrison, D. R. (2017). *E-learning in the 21st century: A community of Inquiry Framework For Research and Practice*. Routledge.
- Garrison, D. R., Anderson, T., & Archer, W. (2001). Critical thinking, cognitive presence, and computer conferencing in distance education. *American Journal of Distance Education*, 15(1), 7–23. <https://doi.org/10.1080/08923640109527071>
- Girón, L. A. M. (2012). *Educating Good citizens: a Case Study of Citizenship Education in Four Multicultural High School Classrooms in Ontario*. University of Ottawa [Unpublished doctoral dissertation].
- Giroux, H. (1978). Writing and critical thinking in the social studies. *Curriculum Inquiry*, 8(4), 291–310. <https://doi.org/10.1080/03626784.1978.11075578>
- Giroux, H. (2014). *Eğitimde kuram ve direniş* (Çev. S. Demiralp). Dost Kitabevi.
- Gonsalvez, L. (2013). *Using Critical Discourse Analysis to Address the gaps, Exclusions and Oversights in Active Citizenship Education*. University of Windsor [Unpublished doctoral dissertation].
- Gürkaynak, İ. (2016). *Düşünen Okul Gelişen öğrenci projesi: Bir Eğitimin Serencamı*. Eğitim Reformu Girişimi.
- Gürkaynak, İ., Gözütok, F. D., Akipek, Ş., Bağlı, M. T., Erhürman, T., & Uluç, F. (2005). *Yurttaş Olmak için... Eğitici El Kitabı*. Umut Vakfı Yayınları.
- International Civic and Citizenship Education Study. (2022). *Assessment framework: international Association for the Evaluation of Educational Achievement (IEA)*. <https://i24.im/MrKTyk2>.
- Johnson, L., & Morris, P. (2010). Toward a framework for critical citizenship education. *The Curriculum Journal*, 21(1), 77–96. <https://doi.org/10.1080/09585170903560444>
- Kansu, C.Ç. (2015). *İlkokul 4. Sınıf öğrencilerinde etkin Yurttaşlık Eğitiminin Etkililiği*. Ondokuz Mayıs Üniversitesi [Unpublished doctoral dissertation].
- Kemmis, S. (1993). Action research. In M. Hammersley (Ed.), *Educational research: Current Issues* (pp. 167–180). Open University.
- Kuhn, D. (1999). A developmental model of critical thinking. *Educational Researcher*, 28(2), 16–46. <https://doi.org/10.2307/1177186>.
- Larson, A. A., Britt, M. A., & Kurby, C. A. (2009). Improving students' evaluation of arguments. *The Journal of Experimental Education*, 77(4), 339–365. <https://doi.org/10.3200/jexe.77.4.339-366>
- Larson, M., Britt, M. A., & Larson, A. A. (2004). Disfluencies in comprehending argumentative texts. *Reading Psychology*, 25(3), 205–224. <https://doi.org/10.1080/02702710490489908>
- Lau, J. Y. F. (2015). Metacognitive education: going beyond critical thinking. In M. Davies, & R. Barnett (Eds.), *The Palgrave handbook of Critical Thinking in Higher Education* (pp. 373–389). Palgrave Macmillan.
- Lipman, M. (2003). *Thinking in Education*. Cambridge University Press.
- Magno, C. (2010). The role of metacognitive skills in developing critical thinking. *Metacognition Learning*, 5(2), 137–156. <https://doi.org/10.1007/s11409-010-9054-4>
- Manalo, E. (2020). Introduction: establishing a case for sharing research-based instructional strategies. In E. Manalo (Ed.), *Deeper learning, Dialogic learning, and Critical thinking: Research-based strategies For the Classroom* (pp. 1–13). Routledge.
- Manalo, E., & Sheppard, C. (2016). How might language affect critical thinking performance? *Thinking Skills and Creativity*, 21, 41–49. <https://doi.org/10.1016/j.tsc.2016.05.005>
- Merriam, S.B. (2015). *Nitel Araştırma Desen Ve Uygulama İçin Bir Rehber* (Çev. S. Turan, Ed.). 3. Baskı, Nobel Yayınları.
- Miles, M. B., & Huberman, A. (1994). *Qualitative Data Analysis*. Sage.
- Murray, D. (1984). Writing and teaching for surprise. *College English*, 46(1), 1–7.
- National Council for the Social Studies. (2001). Creating effective citizens. NCSS position statement. <http://www.socialstudies.org/positions/effectivecitizens>.
- Patton, M.Q. (2014). *Nitel araştırma ve değerlendirme yöntemleri*. (Çev. Ed. M. Büttin, S. Beşir Demir). Pegem Akademi.
- Paul, R., Binker, A. J. A., Martin, D., & Adamson, K. (1989). *Critical Thinking handbook: High school: A guide For Redesigning Instruction*. Center for Critical Thinking and Moral Critique.
- Perkins, D. N. (1987). Thinking frames: an integrative perspective on teaching cognitive skills. J. B. Baron & R. J. Sternberg. *Teaching Thinking skills: Theory and Practice*. W. H. Freeman.
- Phillipson, N., & Wegerif, R. (2020). The thinking together approach to dialogic teaching. In E. Manalo (Ed.), *Deeper learning, Dialogic learning, and Critical thinking: Research-Based Strategies For the Classroom* (pp. 32–47). Routledge.
- Piedade, F., Malafaia, C., Neves, T., Loff, M., & Menezes, I. (2020). Educating critical citizens? Portuguese teachers and students' visions of critical thinking at school. *Thinking Skills and Creativity*, 37, 100690. <https://doi.org/10.1016/j.tsc.2020.100690>.
- Pugliese, G. D. (2015). *Preparing Students For Citizenship in a Global society: A case Study*. Southern Connecticut State University [Unpublished doctoral dissertation].
- Robertson, J. F., & Rane-Szostak, D. (1996). Using dialogues to develop critical thinking skills: a practical approach. *Journal of Adolescent & Adult Literacy*, 39(7), 552–556.
- Siegel, H. (1988). *Educating reason: Rationality, Critical thinking, and Education*. Routledge.
- Sivaci, S., & Altaş, B. (2023). The effect of critical thinking dispositions and democratic values of pre-service EFL teachers on their multicultural self-efficacy: a structural equation modeling approach. *Thinking Skills and Creativity*, 47, Article 101221. <https://doi.org/10.1016/j.tsc.2022.101221>
- Sönmez, B. (2016). *İ. Düşünme Eğitimi Dersinin İlköğretim 6. Sınıf öğrencilerinin eleştirel Ve yaratıcı Düşünme Becerilerine Etkisi*. Anadolu Üniversitesi [Unpublished doctoral dissertation].
- Stanley, T. L. (2017). *Increasing students' Critical Thinking and Improving Performance in Elementary Social Studies Classroom*. Capella University [Unpublished doctoral dissertation].
- Ten Dam, G., & Volman, M. (2004). Critical thinking as a citizenship competence: teaching strategies. *Learning & Instruction*, 14(4), 359–379. <https://doi.org/10.1016/j.learninstruc.2004.01.005>
- The Council of the European Union. (2019). *Key competences for lifelong learning*. <https://op.europa.eu/en/publication-detail/-/publication/297a33c8-a1f3-11e9-9d01-01aa75ed71a1/language-en>.
- Tsui, L. (2002). Fostering critical thinking through effective pedagogy: evidence from four institutional case studies. *Journal-Higher Education*, 73(6), 740–763. <https://doi.org/10.1080/00221546.2002.11777179>
- Van Gelder, T. (2005). Teaching critical thinking: some lessons from cognitive science. *College Teaching*, 53(1), 41–48. <https://doi.org/10.3200/CTCH.53.1.41-48>
- Ververi, O. (2012). *Citizenship Education teachers' Critical Thinking in "education for Democratic citizenship": the sociology of Critical Thinking*. University of Bristol [Unpublished doctoral dissertation].
- Veugelaers, W. (2007). Creating critical-democratic citizenship education: empowering humanity and democracy in Dutch education. *Compare: A Journal of Comparative and International Education*, 37(1), 105–119. <https://doi.org/10.1080/03057920601061893>
- Vijayaratanam, P. (2012). Developing higher order thinking skills and team commitment via group problem solving: a bridge to the real world. *Procedia—Social and Behavioral Science*, 66, 53–63. <https://doi.org/10.1016/j.sbspro.2012.11.247>
- Volman, M., & Ten Dam, G. (2015). Critical thinking for educated citizenship. In M. Davies, & R. Barnett (Eds.), *The Palgrave handbook of Critical Thinking in Higher Education* (pp. 593–603). Palgrave Macmillan.
- World Economic Forum. (2020). *The future of jobs report*. https://www3.weforum.org/docs/WEF_Future_of_Jobs_2020.pdf.
- Yıldırım, A., & Şimşek, A. (2016). *Sosyal Bilimlerde Nitel Araştırma Yöntemleri* (10th ed.). Seçkin Yayınevi.